Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Velocity–time graph

Meaning in my words How would you explain velocity–time graph to a friend?	**How can you tell?** Which axis helps you tell whether the cart is moving left or right?
Example or sketch How would you show three seconds of steady motion at −2 m/s?	**Non-example and why** A student says a flat line at +2 means the cart is parked. What does that height mean?

Constant velocity

Meaning in my words How would you explain constant velocity to a friend?	**How can you tell?** Photographs are one second apart. What pattern would you look for?
Example or sketch Draw three positions for a cart moving 2 m right each second.	**Non-example and why** A bike circles a track at a steady speed. Is its velocity constant?

Frayer models

SECTION A / VOCABULARY Name _____

Find the word on the screen. Talk it through with your partner, then write or sketch in its four boxes.

Signed area

Meaning in my words How would you explain signed area to a friend?	**How can you tell?** What do you do with the area below the time axis when you find displacement?
Example or sketch Sketch a rectangle at −2 m/s lasting 3 s. What displacement does it show?	**Non-example and why** Someone labels the area −2 m/s. Which graph reading did they use?

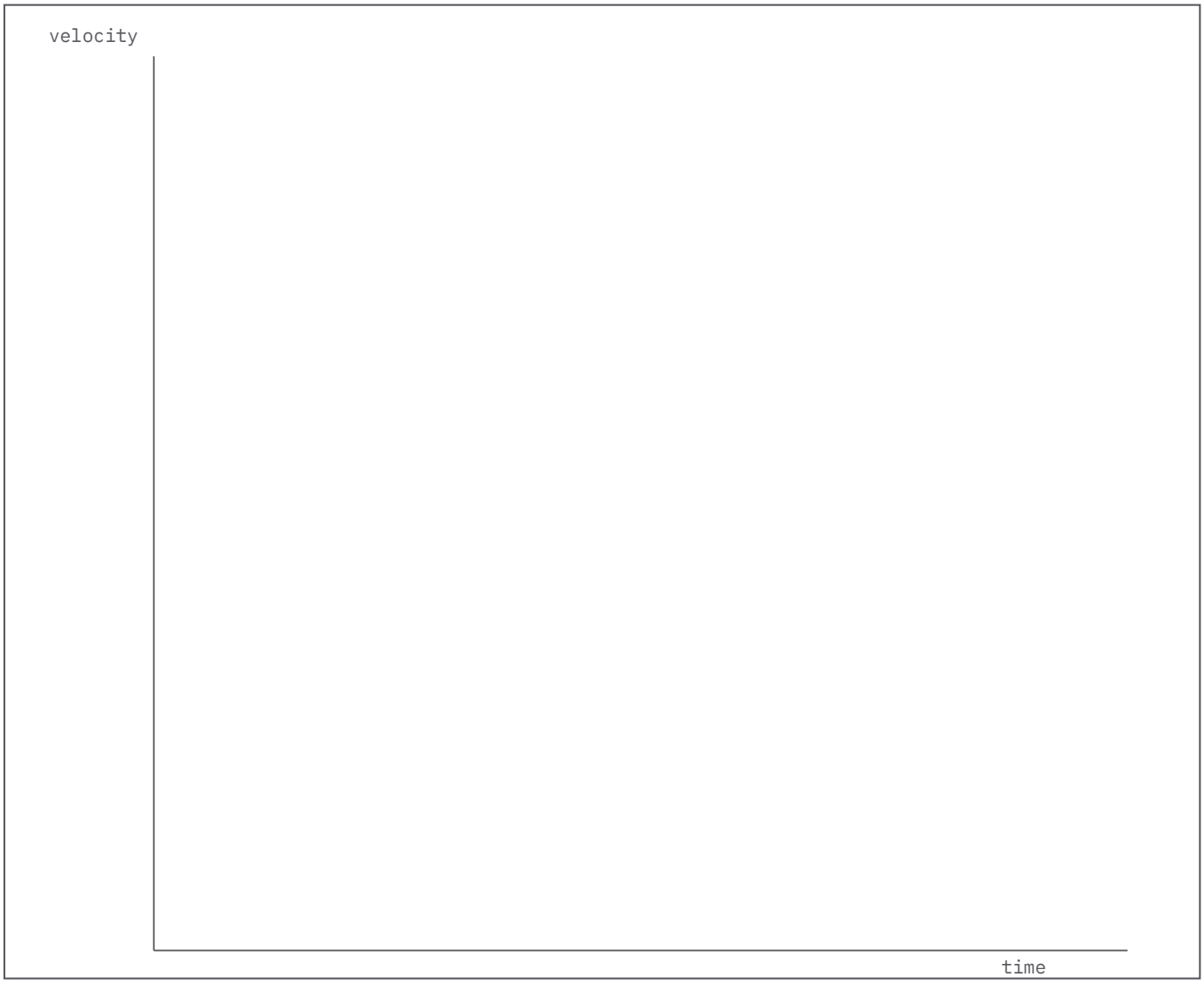
Components

Meaning in my words What are the components of an arrow?	**How can you tell?** If east and north are positive, which component is negative for a southeast trip?
Example or sketch What two arrows could you use to build a displacement of 3 m east and 4 m north?	**Non-example and why** A classmate counts the diagonal and both parts as three separate trips. What is counted twice?

Doodle the idea

SECTION B / DRAW WITH YOUR TEACHER Name _____

Draw and shade opposite-side rectangles. Put m/s on the vertical and s on the horizontal axis.



My marks mean

Where does the camera finish?

04

SECTION E FINAL MEETING / TEAM TRANSFER

Name _____ Partner _____

Our goal: Use signed graph areas to find displacement.

CHOOSE ONE JOB TOGETHER

TRACKING ENGINEER ☐

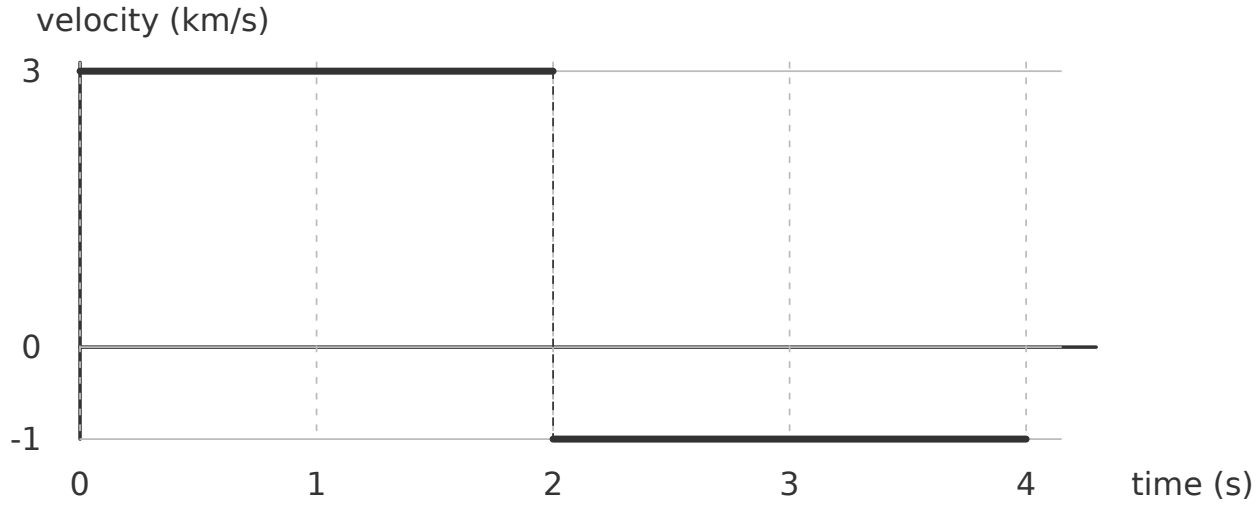
Shade both parts of the velocity graph. Radio the camera's final change in position, including direction.

REPLAY PRODUCER ☐

Mark the outward and return shots. Report total distance and net movement to the editor.

Point to a clue. Keep your first marks; add the update using the color key.

A moving camera follows a model at **+3 km/s** for **2 s**, then **−1 km/s** for **2 s**. The editor needs the end location and the total footage distance.



01 OUR FIRST CALL

Where does the camera finish relative to its start? How far did it travel?

02 OUR REVISED CALL

Wait for the reveal. Keep your first call.

New clue: _____

Check the new caption. Which number belongs on the location graphic? Explain.

Switch speakers. Did you count the return trip with the right sign?

☐ GREEN	☐ RED	☐ YELLOW	☐ BLUE	☐ PURPLE	☐ ORANGE	☐ PENCIL
speeding up	slowing down	steady / balanced	position / displacement	velocity	forces	other marks